

Tangents

Problem 1. Given $\triangle ABC$, let D be the intersection of the line BC with the tangent to the circumcircle Γ of $\triangle ABC$ at A . Let M be the midpoint of AD . Let MB meet Γ again at P and let DP meet Γ again at Q . Prove that $QC \parallel DA$.

Problem 2. In an acute scalene $\triangle ABC$, the altitude from A , the internal angle bisector of $\angle B$ and the internal angle bisector of $\angle C$ bound a triangle with circumcircle Γ . Prove that the internal angle bisector from $\angle A$ is tangent to Γ .

Problem 3. A circle touches the side AB of the triangle ABC at A , touches the side BC at P and intersects the side AC at Q . The line symmetrical to PQ with respect to AC meets the line AP at X . Prove that $PC = CX$.

Problem 4. Let P be a point outside a circle Γ . The tangents from P to Γ touch Γ at the points A and B respectively. Let C be a point on the minor arc AB and let AC meet PB at D . Let E be the point on the segment AB such that $DE \parallel PA$. Prove that $CE \parallel PB$ if and only if $AB^2 = PB \times DB$.

Problem 5. Chords AB and CD of a circle intersect at a point E inside the circle. Let M be an interior point of the segment EB . The tangent line at E to the circle through D , E , and M intersects the lines BC and AC at F and G , respectively. If

$$\frac{AM}{AB} = t,$$

find

$$\frac{EG}{EF}$$

in terms of t .

Problem 6. Let ABC be an acute-angled isosceles triangle with $AB = AC$ and circumcentre O . Lines BO and CO intersect AC , AB respectively at B' , C' . A straight line ℓ is drawn through C' parallel to AC . Prove that the line ℓ is tangent to the circumcircle of $\triangle B'OC$.

Problem 7. Let $ABCD$ be a cyclic quadrilateral and let k_1 and k_2 be circles inscribed in triangles ABC and ABD . Prove that external common tangent of those circles (different from AB) is parallel with CD .

Problem 8. On the hypotenuse AB of a right triangle ABC , we denote a point K such that $BK = BC$. Let P be a point on the perpendicular from the point K to line CK , equidistant from the points K and B . Let L be the midpoint of CK . Prove that line AP is tangent to the circumcircle of $\triangle BLP$.

Problem 9. Given $\triangle ABC$, let D be the intersection of the line BC with the tangent to the circumcircle Γ of $\triangle ABC$ at A . Let M and N be the midpoints of AB and AC respectively. The circumcircle of $\triangle ADM$ meets Γ again at P . The circumcircle of $\triangle ADN$ meets Γ again at Q . Prove that the line PQ bisects AD .

Problem 10. In $\triangle ABC$, let D be a point on the internal angle bisector of $\angle A$. The lines BD and AC meet at E . The lines CD and AB meet at F . The perpendicular bisector of AD meets the line BC at P . Prove that PA is tangent to the circumcircle of $\triangle AFE$.